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ENGINEERING



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SAVEETHA SCHOOL OF ENGINEERING, SIMATS
Department of Computer Science and Engineering
Assignment Information

Particular	Details
Name:	Venganapalli Sireesha
Register Number:	192472208
Department:	CSE(AI)
Programme:	BE-Computer Science and Engineering in Artificial Intelligence
Course Code & Course Name	CSA1709 – Artificial Intelligence
Academic Year / Batch	2026–27
Faculty Name	Dr P Rajaram
Assignment Title	Intelligent Resume Screening and Job Recommendation System
Date of Issue	31.08.2026
Date of Submission	01.09.2026
Maximum Marks	100
Course Outcome(s) – CO	CO2, CO3, CO4 and CO5
Bloom's Taxonomy Level	L3 – Apply, L4 – Analyse, L5 – Evaluate and L6 – Create
SDG Mapping (SDG 1-17)	SDG 4 – Quality Education SDG 8 – Decent Work and Economic Growth SDG 10 – Reduced Inequalities
Industry / Societal Relevance	AI is increasingly used in recruitment to screen resumes, match skills with job requirements and support candidate recommendations. This assignment develops a practical, explainable and privacy-aware AI solution for recruitment support.

Problem Statement:

Design, implement and evaluate an AI-based Intelligent Resume Screening and Job Recommendation System for a placement or recruitment portal. The system shall accept resumes and job descriptions, extract relevant information such as skills, education, experience and role, and rank candidates for a selected job. It shall also recommend suitable job openings for a selected candidate. Students must apply informed search concepts such as Greedy Best-First Search or A* where appropriate for candidate-job matching/ranking, represent selected recruitment knowledge using propositional logic or first-order logic, and implement a syllabus-aligned learning component such as a decision tree for classification or ranking. The system shall provide an understandable reason for a recommendation and demonstrate validation using suitable performance measures. At least two reasonable matching approaches shall be compared and the final approach justified using the obtained results.

Common Assessment Rubric

Assessment Criterion (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Problem Understanding & Formulation (CO2)	10	Clearly defines recruitment problem, users, inputs/outputs, assumptions and constraints; formulation is precise.	Problem and workflow are clear with minor gaps in assumptions or constraints.	Basic problem is identified but formulation is incomplete or unclear.	Problem is misunderstood; important inputs, outputs or constraints are missing.
Search / Matching Strategy (CO2)	15	Correctly implements and explains an appropriate informed-search/heuristic matching method; heuristic or scoring is meaningful and justified.	Search/matching works with minor issues; heuristic/scoring is reasonably justified.	Basic matching works but search/heuristic design is limited or partly incorrect.	Search/matching is missing, inappropriate or non-functional.

Knowledge Representation & Reasoning (CO3)	10	presents domain knowledge by using suitable propositional/FOL reasoning, chaining, resolution or inference, with correct conclusions.	Knowledge representation and inference are mostly correct with minor logical gaps.	Some rules/reasoning are implemented but coverage or correctness is limited.	Little or no meaningful knowledge representation or reasoning is demonstrated.
Learning Model & Recommendation (CO4)	20	Implements a suitable syllabus-aligned learning method; preprocessing, features, training/testing and recommendation logic are correct and clearly explained.	Model and recommendation work correctly with minor issues in features, training or testing.	Basic model is present but learning/recommendation is incomplete or weakly validated.	Model is missing, unsuitable or non-functional.
Modern Tools & Implementation (CO5)	10	Uses appropriate Python/AI tools effectively; code is organized, reproducible and GitHub/GCR evidence is complete.	Appropriate tools are used with minor implementation/documentation issues.	Tools are used at a basic level with limited evidence or organization.	Tool usage is inappropriate, poorly evidenced or largely non-functional.
Results, Testing & Validation (CO4/CO5)	15	Provides clear test cases, metrics, graphs/tables and evidence; results are validated against requirements and	Most tests and metrics are provided; validation is generally correct.	Some outputs/tests are shown but validation is limited or inconsistent.	Insufficient testing, evidence or validation.

		are reproducible.			
Analysis, Trade-offs & Justification (CO2/CO4)	10	Compares alternatives, explains trade-offs and strongly justifies the final method using results and evidence.	Comparison and justification are clear with minor gaps.	Some comparison is given but reasoning is mostly descriptive.	No meaningful comparison or justification.
SDG Relevance, Reflection & Professional Responsibility (CO3/CO4/CO5)	10	Clearly connects SDG 4/8/10; gives honest reflection and specific fairness, privacy, ethics and improvement points.	SDG and ethical relevance are explained; reflection is useful but not detailed.	Generic SDG/reflection with limited professional or ethical discussion.	Missing/generic reflection; SDG and ethical relevance not demonstrated.

Deliverables

To receive full credit, each submission must include the following deliverables:

1. Pseudo code
2. Implementation & Results
3. Github and GCR Upload

INTELLIGENT RESUME SCREENING AND JOB RECOMMENDATION SYSTEM

1. Problem Statement

The Intelligent Resume Screening and Job Recommendation System is an AI-based recruitment support system designed to automatically analyze resumes and job descriptions. The system extracts important information such as skills, education, experience, and job role from resumes and compares them with the requirements of available job openings.

The system ranks candidates for a selected job and recommends suitable jobs for a selected candidate. To achieve intelligent matching, the system uses an informed search technique such as Greedy Best-First Search or A*, knowledge representation using logical rules, and a Decision Tree learning model. The system also provides an understandable reason for each recommendation and evaluates the performance using suitable metrics. At least two matching approaches are compared, and the better-performing approach is selected based on the experimental results.

2. Problem Understanding and Formulation

2.1 Problem

Traditional resume screening requires recruiters to manually examine a large number of resumes. This is time-consuming and may result in inconsistent candidate selection.

The proposed system automates the initial screening process by matching candidate profiles with job requirements using AI techniques.

2.2 Users

The major users of the system are:

- **Recruiters** – to identify suitable candidates.
- **HR departments** – to reduce manual screening effort.
- **Job seekers** – to find suitable job openings.
- **Placement cells** – to recommend jobs to students.

2.3 Inputs

The system accepts:

- Candidate resume
- Candidate skills
- Educational qualification
- Work experience
- Candidate's preferred role
- Job descriptions , Required skills
- Required qualification
- Required experience

2.4 Outputs

The system produces:

- Candidate-job matching score
- Ranked list of candidates
- Recommended jobs
- Eligibility status
- Explanation for recommendation
- Performance evaluation results

2.5 Problem Formulation

Let:

- $C = \{c_1, c_2, \dots, c_n\}$ be the set of candidates.
- $J = \{j_1, j_2, \dots, j_m\}$ be the set of jobs.
- S_c represent candidate skills.
- S_j represent job-required skills.

The objective is to find the best candidate for a selected job:

$$c^* = \arg \max_{c \in C} Match(c, j)$$

Similarly, for a candidate, the system finds:

$$j^* = \arg \max_{j \in J} Match(c, j)$$

The matching function considers skills, education, experience and role compatibility.

3. Objectives

The main objectives are:

1. To develop an AI-based resume screening system.
2. To automatically extract relevant candidate information.
3. To compare candidate skills with job requirements.
4. To rank candidates according to their suitability.
5. To recommend suitable jobs to candidates.
6. To apply informed search for candidate-job matching.
7. To represent recruitment knowledge using logical rules.
8. To implement a Decision Tree learning model.
9. To provide explainable recommendations.
10. To evaluate and compare different matching approaches.

4. Expected Outcomes

The expected outcomes are:

- Faster resume screening.
- Automated candidate ranking.
- Better job recommendations.
- Reduced manual recruitment effort.
- Explainable candidate selection.
- Consistent matching based on defined criteria.
- Quantitative evaluation of the matching system.

5. Requirements

5.1 Functional Requirements

The system should:

- Accept candidate resume information.
- Accept job description information.
- Extract relevant features.
- Calculate candidate-job similarity.
- Rank candidates.
- Recommend jobs.
- Apply logical eligibility rules.
- Apply a learning model.
- Display recommendation explanations.
- Evaluate system performance.

5.2 Technical Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM
- Internet connection for installing required packages

Software

- Python 3.x
- VS Code / Jupyter Notebook
- Scikit-learn
- Pandas , NumPy
- Matplotlib , Git/GitHub

The assignment specifically expects appropriate Python/AI tools and evidence of their usage.

6. Constraints and Assumptions

Constraints

- Resume information must be available in a machine-readable form.
- Matching quality depends on the quality of extracted resume information.
- The training dataset must contain sufficient examples.
- The system should avoid using sensitive personal information for ranking.
- The system should provide understandable recommendations.
- Privacy of candidate information must be maintained.

Assumptions

- Resume information is correctly extracted.
- Job descriptions contain relevant requirements.
- Skills can be represented using standardized terms.
- Candidate experience and education are available.
- The dataset contains sufficient examples for training and testing.

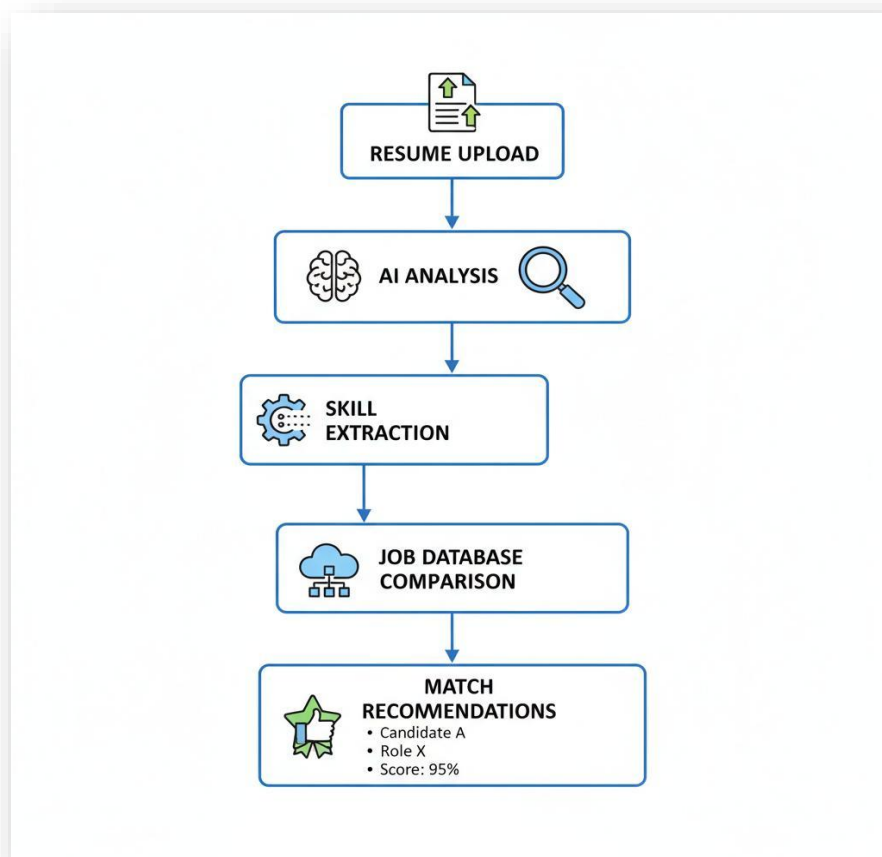
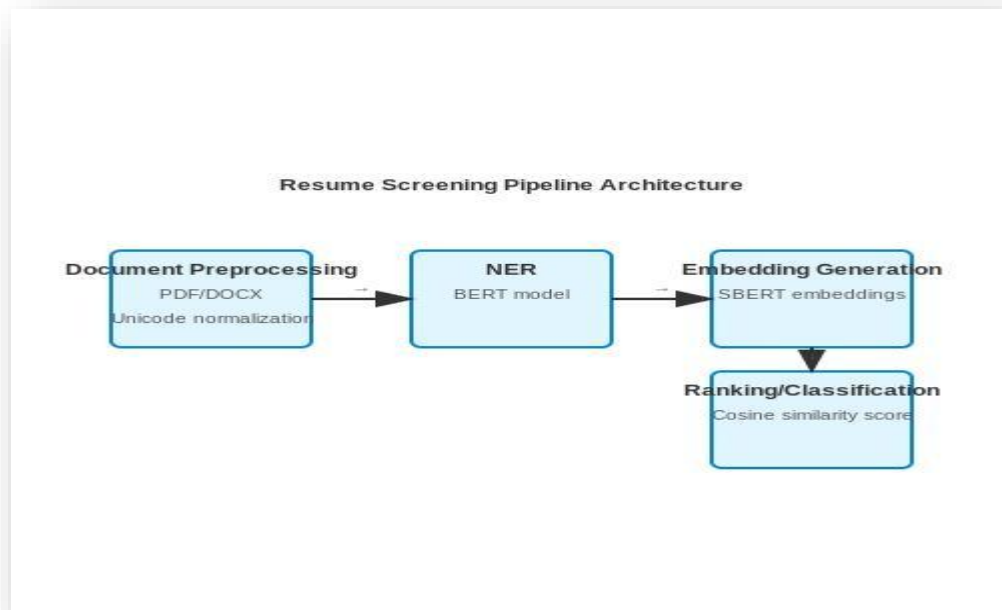
7. Application of AI Concepts

The proposed system applies the following Artificial Intelligence concepts:

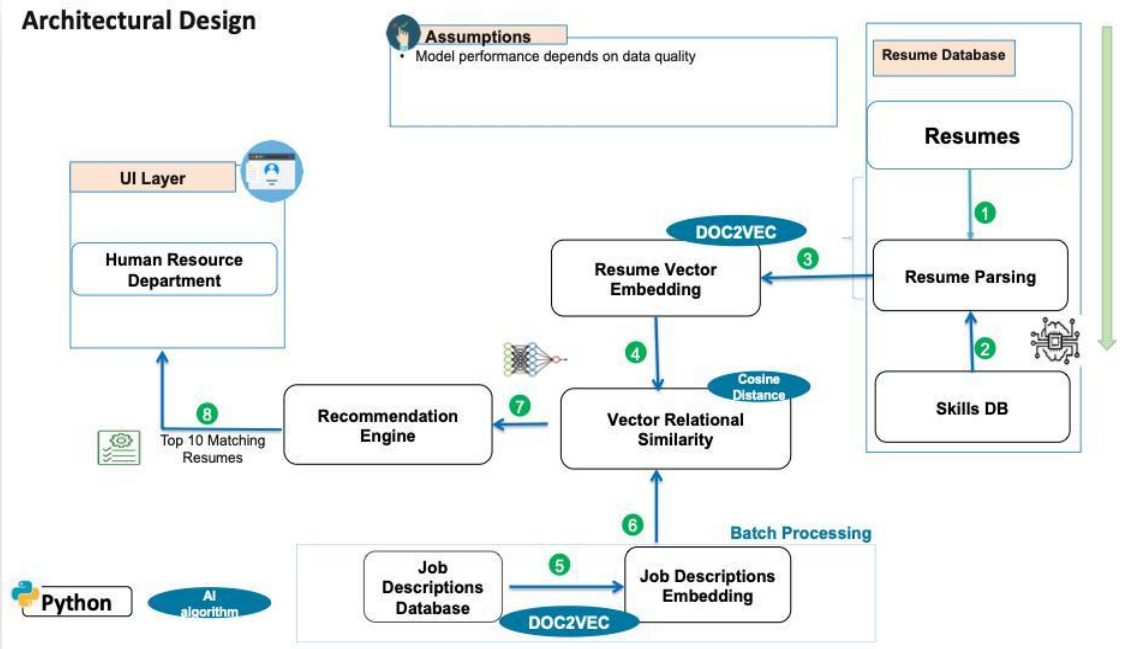
AI Concept	Application
Informed Search	Candidate-job matching
Greedy Best-First Search	Ranking based on heuristic score
A* Search	Cost and suitability-based matching
Knowledge Representation	Recruitment rules
Logical Reasoning	Eligibility determination
Decision Tree	Candidate classification
Feature Engineering	Skills, education and experience
Recommendation System	Job recommendation
Evaluation Metrics	Performance validation

This directly addresses the assignment's requirement to apply course concepts rather than only reproduce theoretical material.

8. Proposed System Architecture



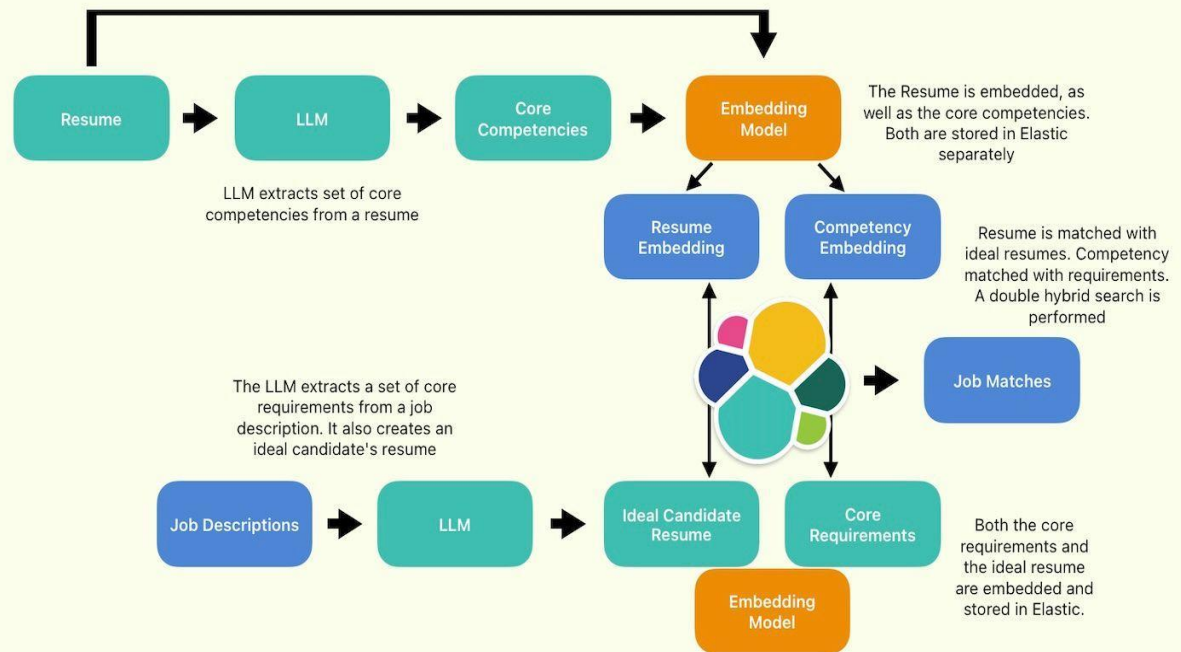
Architectural Design



How AI JD-Resume Matching Works

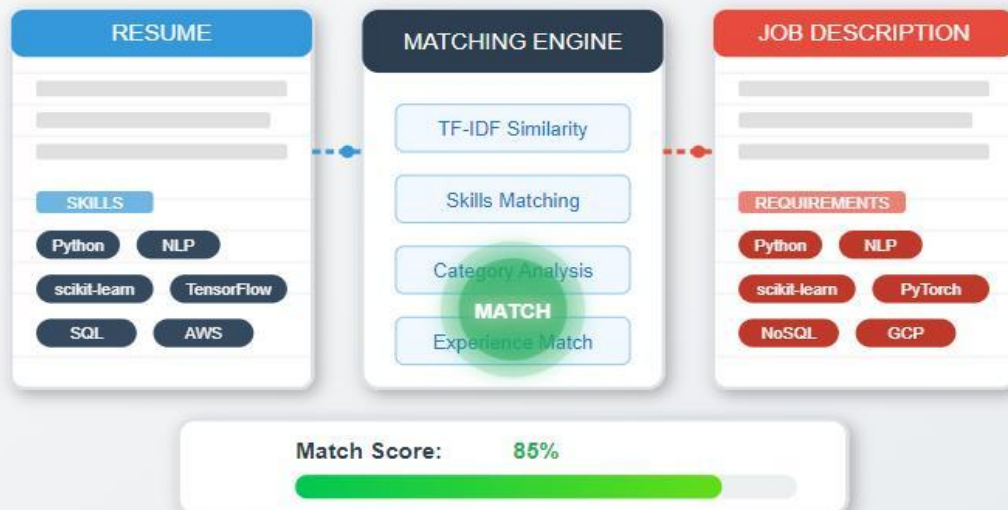


Matching Resumes to Jobs

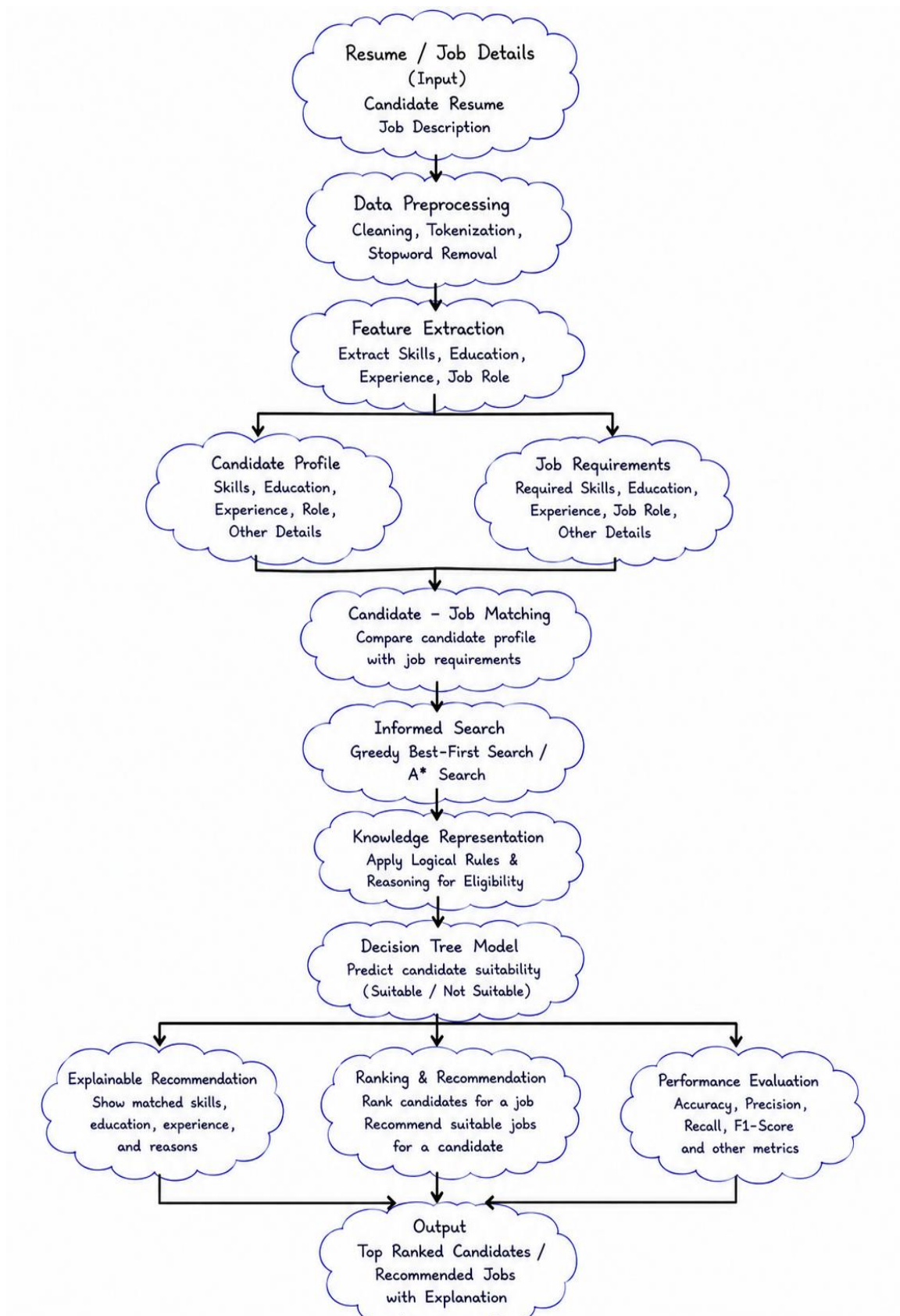


Intelligent Job Matching Agent

Powered by Machine Learning & Natural Language Processing



System Flow



9. Methodology

The system is developed using the following stages:

Step 1: Data Collection

Collect sample resumes and job descriptions.

Step 2: Preprocessing

Clean the input data and convert text into standardized features.

Step 3: Feature Extraction

Extract:

- Skills
- Education
- Experience
- Job role

Step 4: Knowledge Representation

Define recruitment rules such as:

IF candidate has required skill

AND candidate has required education

AND candidate has sufficient experience

THEN candidate is eligible.

Step 5: Matching

Calculate the similarity between candidate features and job requirements.

Step 6: Informed Search

Use Greedy Best-First Search or A* to identify the most suitable candidate/job.

Step 7: Learning Model

Train a Decision Tree using candidate-job features.

Step 8: Recommendation

Generate ranked candidates or recommended jobs.

Step 9: Explanation

Display why a candidate/job was recommended.

Step 10: Evaluation

Evaluate the system using test cases and performance metrics.

10. Feature Representation

For each candidate-job pair, the following features can be generated:

Feature	Description
Skill Match	Percentage of required skills possessed
Education Match	Qualification compatibility
Experience Match	Candidate experience compared with requirement
Role Match	Compatibility of preferred and required role
Overall Score	Combined suitability score

Example:

Skill Match = 80%

Education Match = 100%

Experience Match = 75%

Role Match = 100%

11. Matching Score

A weighted matching score can be defined as: $Score = 0.50S + 0.20E + 0.20X + 0.10R$

Where:

- S = Skill Match
- E = Education Match
- X = Experience Match
- R = Role Match

The weights can be modified according to the recruitment requirement.

For example:

Skill Match = 0.80

Education Match = 1.00

Experience Match = 0.75

Role Match = 1.00

Then: $Score = (0.50)(0.80) + (0.20)(1.00) + (0.20)(0.75) + (0.10)(1.00) = 0.85$

Therefore: $Match\ Score = 85\%$

12. Informed Search – Greedy Best-First Search

Greedy Best-First Search selects the candidate/job that appears most promising according to a heuristic.

The heuristic can be: $h(n) = 1 - MatchScore(n)$

A higher matching score produces a lower heuristic value.

Algorithm

1. Start with all candidate-job pairs.
2. Calculate matching score for every pair.
3. Convert matching score into heuristic value.
4. Insert candidate-job pairs into priority queue.
5. Select the pair with the lowest heuristic value.
6. Continue until the required number of candidates is obtained.
7. Return ranked candidates.

Advantages

- Simple to implement.
- Faster for large candidate lists.
- Uses domain-specific heuristic information.
- Suitable for ranking problems.

The rubric expects the heuristic/scoring mechanism to be meaningful and justified.

13. A* Search

A* can also be used by combining the actual cost and heuristic: $f(n) = g(n) + h(n)$

Where:

- $g(n)$ = cost of reaching the candidate/job state.
- $h(n)$ = estimated remaining cost.
- $f(n)$ = total estimated cost.

For matching: $h(n) = 1 - MatchScore$

The candidate with the lowest $f(n)$ is considered first.

14. Knowledge Representation

Recruitment knowledge can be represented using logical rules.

Facts

has_skill(John, Python)

has_skill(John, Java)

has_degree(John, BTech)

has_experience(John, 2)

Rules

IF candidate has Python

AND candidate has required degree

THEN candidate is technically eligible.

Another rule:

IF candidate has required skills

AND candidate has required experience

THEN candidate is suitable.

Example

has_skill(John, Python)

has_skill(John, SQL)

has_degree(John, BTech)

has_experience(John, 2)

Job requirement:

required_skill(Python)

required_skill(SQL)

required_degree(BTech)

required_experience(1)

Inference:

Python $\rightarrow$ TRUE

SQL $\rightarrow$ TRUE

BTech $\rightarrow$ TRUE

Experience $\geq 1 \rightarrow$ TRUE

Therefore:

John $\rightarrow$ Eligible

This satisfies the rubric requirement for meaningful recruitment knowledge representation and reasoning.

15. Decision Tree Learning Model

A Decision Tree is used to classify a candidate-job pair.

Input Features

- Skill Match
- Education Match
- Experience Match
- Role Match

Target

Suitable = 1

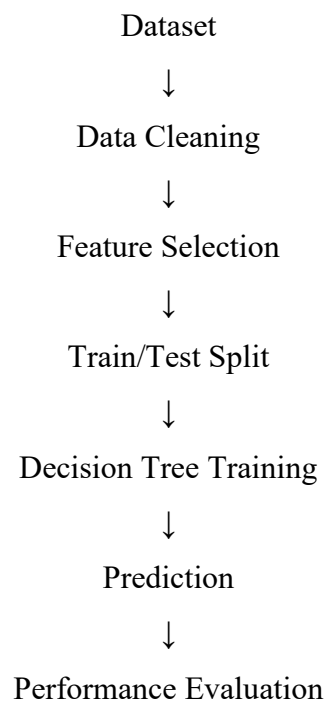
Not Suitable = 0

Example:

Skill	Education	Experience	Role	Suitable
90	100	80	100	1
40	100	30	50	0
80	100	70	100	1
30	50	20	40	0

The Decision Tree learns patterns from these examples and predicts whether a new candidate is suitable.

16. Decision Tree Process



The learning component should clearly demonstrate preprocessing, features, training/testing and recommendation logic, as required for the highest rubric level.

17. Pseudocode

Intelligent Resume Screening and Job Recommendation

BEGIN

Input candidate resumes

Input job descriptions

FOR each resume:

 Extract skills

 Extract education

 Extract experience

 Extract preferred role

Store candidate features

FOR each job:

 Extract required skills

 Extract required education

 Extract required experience

 Extract job role

 Store job features

FOR each candidate-job pair:

 Calculate Skill Match

 Calculate Education Match

 Calculate Experience Match

 Calculate Role Match

 Calculate Overall Match Score

 Apply knowledge-based eligibility rules

 Calculate heuristic value

END FOR

Apply Greedy Best-First Search

Rank candidate-job pairs

Prepare training dataset

Split dataset into training and testing sets

Train Decision Tree

Predict candidate suitability

Calculate evaluation metrics

Generate job recommendations

Generate explanation:

 Show matched skills

 Show education compatibility

 Show experience compatibility

 Show overall score

Display ranked candidates

Display recommended jobs

Display evaluation results

END

18. Explainable Recommendation

The system should not simply display:

Recommended: Candidate A

Instead, it should explain:

Candidate A is recommended because:

- 4 out of 5 required skills matched
- Educational qualification matched
- Required experience satisfied
- Preferred role matched

Overall Match Score: 85%

This makes the AI recommendation understandable to recruiters and candidates.

19. Two Matching Approaches

The assignment requires comparison of at least two reasonable matching approaches.

Approach 1: Weighted Score Matching

The system calculates: $Score = 0.50S + 0.20E + 0.20X + 0.10R$

Approach 2: Informed Search Matching

Greedy Best-First Search uses the matching score as a heuristic and prioritizes the most promising candidate-job pair.

20. Comparison

Parameter	Weighted Matching	Greedy Best-First Search
Implementation	Simple	Moderate
Ranking	Yes	Yes
Heuristic	Limited	Explicit
Search concept	No	Yes
Computational complexity	Low	Moderate
Explainability	High	High
AI search requirement	Limited	Strong
Suitability for assignment	Good	Very Good

Final Selection

Greedy Best-First Search combined with weighted matching is selected as the final approach because it provides both a meaningful matching score and an informed search mechanism. It also directly demonstrates the AI search concept required by the assignment.

The final decision should ultimately be supported by the actual experimental results rather than only descriptive reasoning. The rubric specifically expects alternatives, trade-offs and evidence-based justification.

21. Test Cases

Test Case	Input	Expected Result
TC01	Candidate has all required skills	High match
TC02	Candidate missing several skills	Low match
TC03	Education matches requirement	Eligible
TC04	Experience below requirement	Lower ranking
TC05	All requirements satisfied	Recommended
TC06	Candidate role does not match	Lower ranking
TC07	No matching jobs	No suitable recommendation

22. Sample Results

Example:

Candidate	Skill Match	Education	Experience	Role	Overall
Candidate A	90%	100%	80%	100%	91%
Candidate B	70%	100%	60%	80%	74%
Candidate C	50%	100%	40%	60%	58%

Ranking

1. Candidate A – 91%
2. Candidate B – 74%
3. Candidate C – 58%

Therefore, Candidate A is the highest-ranked candidate.

Note: These are sample values for demonstrating the report format. Replace them with your actual program outputs before submission.

23. Performance Metrics

The following metrics can be used:

$$\textbf{Accuracy : } Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

$$\textbf{Precision : } Precision = \frac{TP}{TP+FP}$$

$$\textbf{Recall : } Recall = \frac{TP}{TP+FN}$$

$$\textbf{F1 Score : } F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

These metrics help determine whether the learning model correctly identifies suitable and unsuitable candidates.

The rubric expects test cases, metrics, tables/graphs and evidence that the results satisfy the requirements.

24. Results and Validation

The system is validated using multiple candidate-job combinations.

Validation includes:

- Correctness of skill matching.
- Correctness of eligibility rules.
- Candidate ranking.
- Job recommendation.
- Decision Tree predictions.
- Precision.
- Recall.
- F1-score.
- Comparison between matching approaches.

The obtained results should be compared with expected results to identify incorrect classifications and ranking errors.

25. Modern Tools Used

Tool	Purpose
Python	Main programming language
Pandas	Data processing
NumPy	Numerical operations
Scikit-learn	Decision Tree and metrics
Matplotlib	Result visualization
VS Code	Development
Git	Version control
GitHub	Source-code submission

The assignment requires appropriate modern tools and evidence of their usage.

26. Analysis and Trade-offs

Weighted Matching

Advantages:

- Easy to understand.
- Fast.
- Easy to customize.
- Highly explainable.

Limitations:

- Depends on manually selected weights.
- May not capture complex relationships.

Greedy Best-First Search

Advantages:

- Uses heuristic information.
- Efficient for prioritizing promising candidates.
- Demonstrates informed search.

Limitations:

- Greedy selection may not always produce a globally optimal result.
- Performance depends on the quality of the heuristic.

Decision Tree

Advantages:

- Easy to interpret.
- Handles numerical and categorical features.
- Provides understandable decisions.

Limitations:

- Can overfit.
- Performance depends on training data quality.

Final Decision

A hybrid approach combining weighted matching + informed search + Decision Tree provides a stronger solution because each component performs a different function:

Weighted Score → measures suitability



Informed Search → ranks efficiently



Decision Tree → learns suitability patterns



Knowledge Rules → checks eligibility



Explanation → communicates the reason

27. SDG Relevance

The assignment specifically maps the project to SDG 4, SDG 8 and SDG 10.

SDG 4 – Quality Education

The system can help students and graduates identify job opportunities based on their skills and qualifications. It can also help placement cells recommend suitable employment opportunities.

SDG 8 – Decent Work and Economic Growth

The system improves the connection between qualified candidates and employment opportunities. Faster and more systematic recruitment can contribute to efficient employment processes.

SDG 10 – Reduced Inequalities

A consistent skill-based screening process can reduce some forms of arbitrary selection. The system should avoid using irrelevant sensitive attributes during ranking.

28. Ethical Considerations

Fairness

The system should evaluate candidates primarily using job-relevant qualifications and skills.

Privacy

Resume information contains personal data and should be securely stored and processed.

Bias

Training data may contain historical recruitment biases. Therefore, the model should be evaluated for unfair patterns.

Transparency

The system should explain why a candidate was recommended.

Human Oversight

The AI system should support recruiters rather than completely replace human decision-making. The rubric explicitly expects discussion of fairness, privacy and ethics along with SDG relevance.

29. Limitations

The proposed system has the following limitations:

1. Resume formats may differ significantly.
2. Skill extraction may not always be accurate.
3. Synonyms for skills can affect matching.
4. The Decision Tree depends on training data.
5. Historical data may contain bias.
6. Greedy search may not always find the globally optimal ranking.
7. The system does not completely replace human recruitment decisions.

30. Future Enhancements

Future versions can include:

- Natural Language Processing for better resume parsing.
- Transformer-based resume understanding.
- Semantic similarity using embeddings.
- Larger real-world datasets.
- Multilingual resume processing.
- Advanced fairness evaluation.
- Cloud-based deployment.
- Secure database integration.
- Interactive recruiter dashboard.
- Personalized career recommendations.

31. Conclusion

The Intelligent Resume Screening and Job Recommendation System provides an AI-based solution for automating candidate screening and job recommendation. The system analyzes skills, education, experience and job roles to calculate candidate-job suitability.

The proposed approach combines weighted matching, informed search, knowledge representation and Decision Tree learning. The system ranks candidates, recommends suitable jobs and provides understandable explanations for its decisions.

The comparison of matching approaches helps identify the strengths and limitations of each method. With proper testing, validation, privacy protection and fairness considerations, the system can serve as an effective recruitment-support tool.

32. Individual Contribution

You can modify this according to your actual team members.

Team Member	Contribution
Member 1	Problem formulation, data preprocessing and resume feature extraction
Member 2	Matching algorithm, informed search and knowledge representation
Member 3	Decision Tree model, testing, evaluation and documentation

Important: The assignment instructions state that all members must actively participate and individual contributions must be clearly specified.

33. Individual Reflection

Individual Reflection

This assignment helped me understand how Artificial Intelligence concepts can be applied to a practical recruitment problem. I learned how a resume screening system can use candidate skills, education, experience and job role as features for matching candidates with suitable job opportunities. Instead of manually comparing every resume with a job description, an AI-based system can automate the initial screening and ranking process.

One of the important concepts I learned through this assignment was informed search. I understood how a heuristic can be designed using a candidate-job matching score and how Greedy Best-First Search can be used to prioritize promising candidates. I also learned the importance of knowledge representation and reasoning. By defining recruitment rules, the system can determine whether a candidate satisfies important eligibility conditions.

I also gained practical knowledge about Decision Tree classification. The process of preparing features, splitting data into training and testing sets, training the model and evaluating predictions helped me understand the application of machine learning beyond theoretical concepts. Comparing different matching approaches also helped me understand that an AI solution should not be selected only because it is easy to implement; its performance, explainability and limitations should also be considered.

A major challenge was deciding which candidate attributes should influence the matching score and how much importance should be given to skills, education, experience and role. Another challenge was ensuring that the recommendation is understandable instead of producing only a numerical score. This helped me understand the importance of explainable AI in real-world applications.

The assignment is also related to SDG 4, SDG 8 and SDG 10 because the system can support education-to-employment opportunities, improve employment matching and encourage more consistent skill-based screening. I also learned that recruitment AI must consider privacy, fairness and bias because incorrect or biased recommendations can negatively affect candidates.

If additional time and resources were available, I would improve the system using more realistic resume datasets, semantic NLP techniques and a larger evaluation dataset. I would also add stronger fairness and privacy checks. Overall, this assignment improved my understanding of AI search, knowledge representation, machine learning, recommendation systems and responsible AI development.

34. References

Use references for all external datasets, software, algorithms and other sources used. The assignment format specifically requires external sources, datasets, software and AI-assisted tools to be acknowledged.

Suggested references:

1. Li, C., Fisher, E., Thomas, R., Pittard, S., Hertzberg, V., & Choi, J. D. (2020). Competence-level prediction and resume-job description matching using transformer models. *Proceedings of EMNLP 2020*.
2. Malinowski, J., Keim, T., Wendt, O., & Weitzel, T. (2020). A machine learning approach for automation of resume recommendation systems. *Procedia Computer Science*, 167, 2318–2327.
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4. Köchling, A., & Wehner, M. C. (2022). Discriminated by an algorithm: A systematic review of discrimination and fairness in algorithmic decision-making in HR recruitment. *Business Research*, 15, 795–848.
5. Li, C., Huang, Y., et al. (2023). Talent recommendation based on attentive deep neural networks and implicit relationships of resumes. *Information Processing & Management*, 60(4), 103357.
6. Pimpalkar, A., Lalwani, A., Chaudhari, R., Inshall, M., Dalwani, M., & Saluja, T. (2023). Job application selection and identification using natural language processing and machine learning. 2023 IEEE International Students' Conference on Electrical, Electronics and Computer Science.

7. Wilson, K., & Caliskan, A. (2024). Gender, race, and intersectional bias in resume screening via language model retrieval. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*.
8. Abhishek, K. L., Niranjanamurthy, M., Aric, S., Ansarullah, S. I., Sinha, A., Tejani, G., & Shah, M. A. (2025). Developing an intelligent resume screening tool with AI-driven analysis and recommendation features. *Applied AI Letters*, 6(2), e116.
9. Kumar, D., Verma, C., & Illés, Z. (2026). Optimizing student job placements with NLP and explainable AI: A fair and transparent hiring framework. *Array*, 29, 100729.
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